

Exploratory Data Analysis (EDA) & Business Intelligence

Data Analytics Internship — Task 2

ApexPlanet Pipeline Execution

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Exploratory Data Analysis (EDA) & Business Intelligence

Transforming 1,000 raw sales records into actionable business insights.

| Core Task Objectives



Descriptive Statistics

Calculating summaries and trends to understand the data's fundamental distribution and layout.



SQL Aggregations

Executing targeted data queries to pull out vital revenue markers and answers directly from the dataset.



KPI Mock-up

Designing a static dashboard framework to structure critical baseline metrics cleanly for leadership review.

| Dataset Integrity Summary

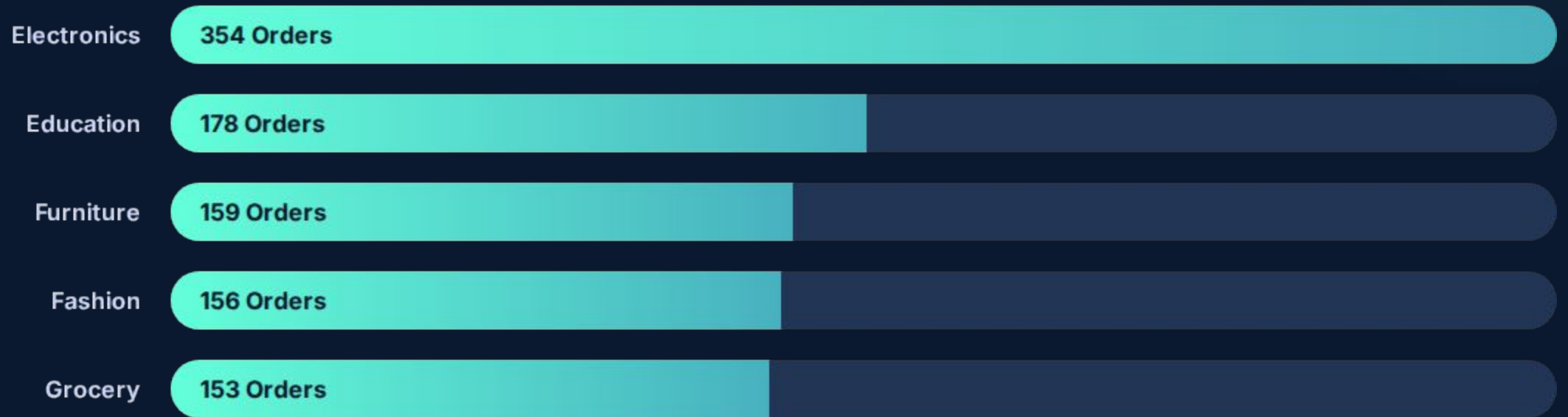
1,000

Processed Transactions

Data Posture Check

Following the initialization steps, the data files preserve strict column accuracy. Missing entry vectors across fields like `Age` and `City` have been handled cleanly, paving the way for valid correlation models.

| Univariate Analysis: Category Lead



*****Key Insight:**** Electronics holds the clear volume advantage, mapping out over ****35%**** of all recorded user trends.*

| SQL Business Intelligence

`</>`Query: Identifying Top Revenue Drivers.

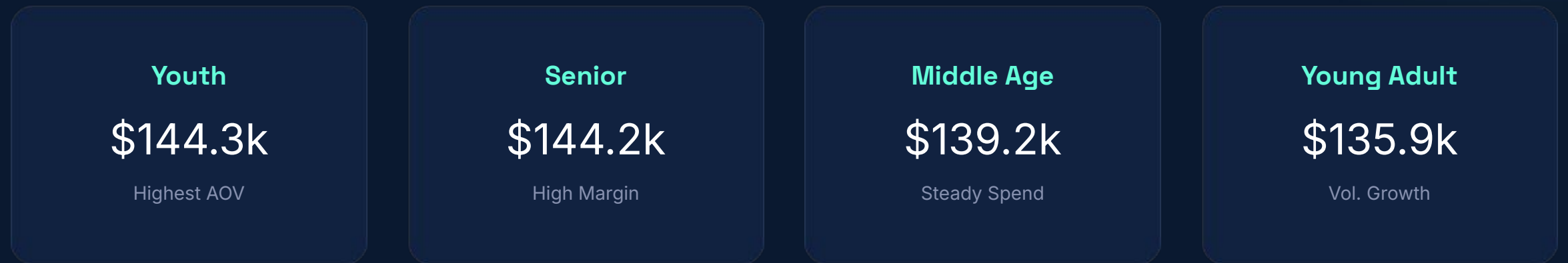
```
SELECT Product, Category, SUM(Total_Sales)
FROM sales_data
GROUP BY 1, 2
ORDER BY 3 DESC LIMIT 5;
```

✓ **Laptop:** \$25.44M Revenue

✓ **Mobile:** \$25.33M Revenue



| Average Order Value (AOV) Insights



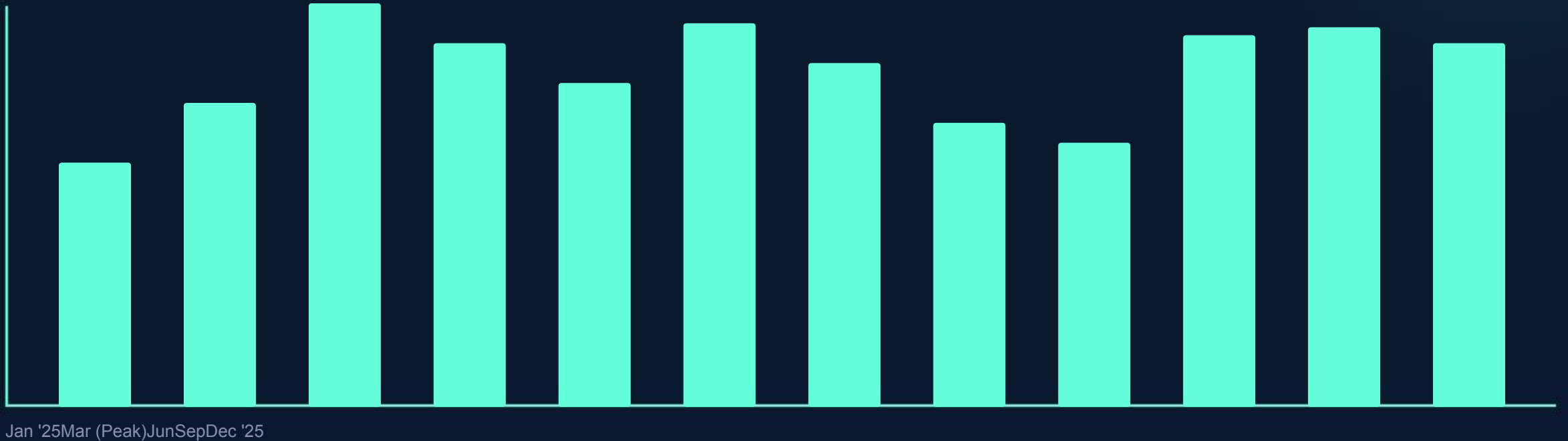
The analysis reveals that ****demographic extremes (Youth & Seniors)**** actually contribute the highest value per transaction, identifying a key opportunity for premium marketing targeting.

| VIP Customer Spotlight (LTV)

Rank	Customer Name	ID	Lifetime Value (\$)
1	Customer_254	CUST2062	\$493,677.50
2	Customer_200	CUST4706	\$492,174.10
3	Customer_36	CUST1711	\$490,866.40

These high-value individuals represent our ****Tier 1 Loyalists****. Retaining these customers is critical for maintaining high annual revenue ceilings.

Monthly Revenue Trends



****March 2025**** peaked at ****\$13.06M****, showing a significant seasonal spike in purchasing activity.

| Static Dashboard Proposal

TOTAL REVENUE

\$159.4M

TOTAL ORDERS

1,000

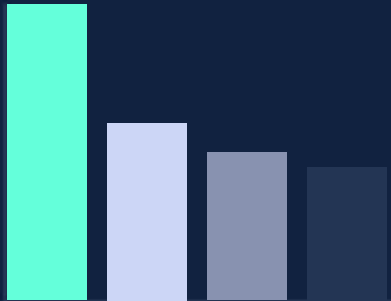
AVG BASKET

\$159k

TOP REGION

Patna

Revenue by Category



Regional Share



This dashboard mock-up represents the critical KPIs proposed for tracking sales health moving forward.

Any Questions?

Exploratory Data Analysis (EDA) & Business Intelligence — Task 2 Complete.



<https://github.com/fatimahadnan230/ApexPlanet-Data-Analytics-Internship/tree/main>



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